Using X11 with Engaging Cluster

While using the command line is often fast and efficient, we often just need to use a Graphical User Interface (GUI).

This process allows you to use the cluster and present a GUI to your desktop machine using X11.

Mac and Linux

On Linux or Mac, Open the terminal program of your choice. On the Mac, you will have to make sure you have XQuartz installed and you will need to use either XQuartz or iTerm2 as your terminal.

ssh to the cluster log in node using your MIT kerberos ID.

```
ssh -X <your_mit_kerb>@orcd-login.mit.edu
```

You will get a DUO push 2 factor authentication notification asking you to log in.

Once you are logged in, create your allocation in slurm using X11 option:

```
salloc -N 1 -n 4 --mem-per-cpu=4G -p mit_normal --x11
```

Load the module you want to use a GUI with. In this example, we're going run MATLAB on the cluster and have it display on our desktop or laptop:

```
module load matlab

matlab &
```

The "&" in the above command puts the matlab program in the background and frees your terminal for other activities.

Note

If you are on a wireless network, the GUI may not show up on your desktop for several seconds. If you are connected via ethernet cable, it will be more responsive. For example, working remotely at the Rotch library on wireless it could take 20 seconds for the GUI to display on your system. This is not likely something you would want to do for your regular workflow, but it can be useful for small tasks or if you do not have another option.

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